

# Topology-Aware & Hierarchical All-Reduce; Libraries

- Libraries:
  - NCCL (GPU-centric), Gloo (CPU/GPU), MPI (HPC)
- Algorithms:
  - Ring: bandwidth-optimal for large messages
  - Tree: lower latency for smaller messages
  - Reduce-scatter + all-gather: core of sharded training
- Hierarchical all-reduce:
  - Intra-node reduce-scatter (NVLink)
  - Inter-node reduce-scatter (IB/Ethernet)
  - Inter-node all-gather
  - Intra-node all-gather
- Practice: bucket sizes, algorithm selection per tensor size, topology hints

# Tensor Parallelism & Sharding (Megatron, ZeRO/FSDP)

## Tensor parallel (intra-layer):

- Split attention heads or MLP dimensions across GPUs
- Per-layer collectives: all-gather/all-reduce or reduce-scatter in fwd/bwd
- Works best within fast NVLink/NVSwitch “islands”

## Optimizer/gradient/parameter sharding:

- Stage 1: shard optimizer states
- Stage 2: shard optimizer + gradients
- Stage 3: shard optimizer + gradients + parameters (optionally offload to CPU/NVMe)

Reduces per-rank memory; introduces extra comm for gathers/scatters

## FSDP (PyTorch):

- Fully sharded per module; parameters/optimizer states are sharded
- All-gather param shards before compute; reduce-scatter grads after
- Configurable wrap/granularity, mixed precision, and offload

## Trade-off:

- Big memory relief enables very large models
- More frequent and fine-grained communication—tune bucket sizes, prefetch, and overlap
- Combine with activation checkpointing and BF16/FP16

# Numerical Precision in AI Hardware

Trading Accuracy for Efficiency

## Precision Format Evolution:

```
// Training: High precision for gradient stability
float32 weights[1024][512]; // 32-bit floating point
float32 activations[batch][1024];

// Inference: Reduced precision for efficiency
int8_t quantized_weights[1024][512]; // 8-bit integer
int8_t quantized_activations[batch][1024];

// Ultra-low power: Extreme quantization
int4_t binary_weights[1024][512]; // 4-bit or binary
```

### Hardware Implications:

- **FP32**: Full accuracy, high energy cost, large storage
- **FP16/BF16**: 2x memory reduction, faster computation on modern hardware
- **INT8**: 4x memory reduction, specialized integer units, minimal accuracy loss
- **INT4/Binary**: Extreme efficiency, requires careful algorithm design

### Mixed-Precision Computing:

Modern accelerators support different precisions for different operations:

- Matrix multiplication: INT8 or FP16
- Accumulation: FP32 for numerical stability
- Activation functions: INT8 for efficiency